

Lucas Thiessen

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Education

University of Waterloo

September 2020 – April 2025

Bachelor of Applied Science in Computer Engineering

Skills

Programming Languages: C++/C, Verilog, VHDL, RISC-V, Python, Java, Kotlin, PHP, SQL, JavaScript, HTML/CSS

Tools: UNIX/Linux, Git, Apache, PyTorch, OpenCV, NumPy, PCB Design, DMM, PuTTY, UART, STM32, Figma

Experience

Software Developer | Mikobyte Solutions

January 2024 – Present

- Refactored the company's core **C++ networking stack** (TCP connections, socket abstractions, thread pools, **IOCP**-based I/O, multicast support), validated with test applications and integrated into trading and blockchain systems.
- Built **cryptocurrency node software** with custom P2P networking, consensus rules (mempool, longest-chain selection, mining logic), multi-wallet RPC support, and persistent storage using **LevelDB/SQLite**.
- Implemented a **C++ market simulator and trading client** with internal networking and **FIX-style protocols**, featuring an order book, matching engine, price simulation, and desktop UI for order entry and market monitoring.
- Built **MFC-based developer tooling** to scaffold full-stack applications, generating backend services, static sites, and React frontends, along with database schemas and optional AI-assisted branding and theming.

Software Engineer | AlertDriving

January 2023 – April 2023

- Developed and deployed over 20 software enhancements and bug fixes using **PHP, JavaScript, and SQL**, improving platform functionality and user experience.
- Automated employee onboarding workflows by implementing bulk account creation, Excel import processing, and automated account setup notifications.

Back-End Software Developer | Year Zero Studios

May 2022 – August 2022

- Developed a **React and Firestore-based** quiz authoring platform, improving content creation workflows and overall user experience.
- Delivered features across multiple client projects using **React, Firestore, Figma, and Heroku** while collaborating within an agile development environment.

Computing Assistant | University of Waterloo

September 2021 – December 2021

- Maintained university computing infrastructure and asset-management systems, resolving technical issues and performing hardware upgrades to ensure reliable operation.

Projects

CargoBuddy: Advanced Autonomous Cargo Delivery System

May 2024 – April 2025

- Developed an autonomous cargo-delivery robot using **C++** on Raspberry Pi, implementing path-planning and particle-filter localization using **LiDAR** and **ultrasonic sensors**.
- Designed **PCB** hardware and integrated sensing, navigation, and control systems for autonomous operation.

RTL 5-Stage Pipelined Processor

September 2023 – December 2023

- Designed a 5-stage pipelined **RISC** processor on a **PYNQ-Z1 FPGA** using **Verilog/SystemVerilog**, implementing hazard detection, forwarding logic, and timing-verified execution.

Altera Cyclone FPGA Operating System

May 2023 – August 2023

- Developed operating-system components in **C** including first-fit memory management, priority scheduling, task aging, and inter-process communication mechanisms.